

Sahil Raj Patel

AI Automation Engineer | Generative AI | Agentic AI | LLM Applications
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PROFILE

AI-focused B.Tech Computer Science undergraduate specializing in **Generative AI, LLM applications, Agentic AI, and intelligent automation**. Hands-on experience building **RAG systems, AI agents, automation workflows, and machine learning applications** using Python, LangChain, LangGraph, Hugging Face, OpenAI APIs, and modern automation frameworks.

TECHNICAL SKILLS

Languages: Python, C, SQL

Generative AI & LLMs: LLMs, RAG, PEFT, LoRA, QLoRA, Fine-Tuning, Hugging Face, Transformers, NLP, Prompt Engineering

Agentic AI & Frameworks: AI Agents, Agentic Workflows, LangChain, LangGraph, LlamaIndex, Tool/Function Calling, Multi-Agent Systems

Vector Search & Databases: Semantic Search, Text Embeddings, Pinecone, PostgreSQL, SQL

AI Automation & APIs: n8n, OpenAI API, Gemini API, Workflow Automation, MLOps

ML / DL & Computer Vision: Deep Learning, Computer Vision, CNN, OpenCV, YOLOv8, PyTorch, TensorFlow, Scikit-learn

Tools: Pandas, NumPy, Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Docker, Power BI, Microsoft Excel, VBA

EXPERIENCE

Tata Motors India

AI / Data Science Summer Intern

2026

Jamshedpur, India

- Developed **Digitization of Inventory**, an automation solution for the engine manufacturing unit to streamline daily part-shortage tracking, reporting, clearance analysis, and buyer communication.
- Built **Outlook VBA automation** to automatically retrieve and save daily shortage-report Excel attachments, followed by centralized data consolidation into a master Matrix workbook.
- Automated shortage mapping and **engine clearance calculation** using Excel VBA, SUMIF-based logic, and Bill of Materials (BoM) constraints; generated shortage insights for production planning.
- Developed a **Power BI shortage dashboard** to visualize shortages by part, buyer, vendor, and engine model, and implemented automated VBA-based email notifications for responsible buyers.

CertED Technologies, Haridwar University

Data Science Summer Intern

June – July 2025

Haridwar, India

- Built and evaluated machine learning models achieving **90%+ accuracy**; improved model performance by **12%** using cross-validation and hyperparameter tuning.
- Performed **data preprocessing, feature engineering, and model evaluation**; deployed machine learning models using **Flask REST APIs and Docker**.

PROJECTS

RAG Application

Python · LangChain · LangGraph · OpenAI · Pinecone

[GitHub](#)

- Developed a document-grounded **Retrieval-Augmented Generation (RAG)** application supporting PDF and text ingestion, document chunking, embedding generation, semantic retrieval, and context-aware answer generation.
- Designed a **LangGraph retrieval-to-generation workflow** with separate retrieval and generation nodes for modular and grounded response generation.

AI Automation Agent

Python · LangChain · LangGraph · Playwright

[GitHub](#)

- Built an **agentic browser automation system** with reusable tools for website navigation, search, YouTube actions, WhatsApp messaging, and automated form filling.
- Implemented a **persistent browser context** with LangGraph orchestration to maintain browser state and execute multi-step tasks.

AI-Based Autonomous Face Tracking & Target Locking System

Python · OpenCV · YOLOv8 · Face Recognition · Computer Vision

[GitHub](#)

- Won **2nd Position at TECHSANGRAM 2025** by developing a real-time autonomous face recognition and target-tracking system operating at approximately **30 FPS**.
- Integrated face recognition with **YOLOv8-based detection and tracking**, achieving **94% face recognition accuracy** and reducing false detections by **15%** through preprocessing and model optimization.

EDUCATION

Haridwar University <i>B.Tech in Computer Science and Engineering – 76%</i>	2023 – 2027 <i>Haridwar, India</i>
Premalok Mission School <i>Senior Secondary (CBSE) – 80%</i>	<i>India</i>
<i>Secondary (CBSE) – 82%</i>	<i>India</i>

CERTIFICATIONS

Data Science Summer Internship – CertED (2025) IBM SkillsBuild C Programming – IIT Bombay Spoken Tutorial (95%)	Data Science with Generative AI – PW Skills Microsoft Learn – AI Internship / Power BI NPTEL Certification
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